འཇིགས་མེད་རྣམ་རྒྱལ་བཙན་རིག་མཐོ་རིམ་སློབ་གྲྭ།

Royal University of Bhutan

Jigme Namgyal Engineering College, Dewathang

Report on Network Design and Implementation Using BGP

(A multihomed hospital network using eBGP and Access Control Lists, built in Cisco Packet Tracer)

Submitted by

Sherab Pem(05250188)

DIPLOMA IN COMPUTER SYSTEM AND NETWORK

DEPARTMENT OF INFORMATION TECHNOLOGY

DATE 15/10/2026

Contents

1. Introduction and Scenario 3
2. Purpose and Requirements 3
3. Network Topology and Devices 3
4. IP Addressing and AS Plan 6
4.1 Autonomous systems 6
4.2 Router links 6
4.3 LANs and end devices 6
4.4 End-device configuration 6
5. Router Configuration 8
5.1 Hospital-R (hospital edge router) 8
5.2 Bhutan Telecom (Router6) 9
5.3 Tashicell (Router8) 10
5.4 Link connectivity check 11
6. BGP and eBGP Implementation 13
6.1 How BGP is used here 13
6.2 Redundancy and path selection 13
7. ACL Implementation 13
7.1 PROTECT-DB: protecting the database 13
7.2 Why outbound on Gi0/2 14
7.3 Limitation of PROTECT-DB 14
8. Testing and Verification 15
8.1 Hospital-R 15
8.2 Bhutan Telecom (Router6) 19
8.3 Tashicell (Router8) 20
8.4 Connectivity and ACL tests 20
8.5 Summary of results 25
9. Conclusion 26
full Configuration 27
A.1 Hospital-R 27
A.2 Bhutan Telecom (Router6) 27
A.3 Tashicell (Router8) 28

1. Introduction and Scenario

The Jigme Dorji Wangchuck National Referral Hospital (JDWNRH) in Thimphu is Bhutan's largest hospital. It runs an online appointment and information portal for patients and the public. Its internal network holds the public web server, the database of confidential patient records and the computers used by hospital staff. Only the web server should be reachable from the internet. The patient database must stay hidden.

The hospital buys internet access from two providers, **Bhutan Telecom (BT)** and **Tashicell**. Using two providers is called multihoming. It keeps the appointment portal online if one provider fails. The hospital, Bhutan Telecom and Tashicell are three separate organisations with their own routing policies. Each therefore has its own Autonomous System (AS), and the links between them run **eBGP**.

The hospital also needs to protect patient privacy. An **extended ACL** on the hospital router denies all access to the database server from outside the hospital, while the web portal stays available.

This is a lab simulation for the assignment. The AS numbers are private (65001, 65010 and 65020) and the addresses are private lab addresses. The design does not represent the real configuration of any of these organisations.

2. Purpose and Requirements

The network must meet the following requirements:

- Connect the hospital, Bhutan Telecom and Tashicell with eBGP, one AS per organisation.
- Advertise each organisation's LAN to the other two through BGP.
- Keep a backup path: if one link fails, traffic uses the remaining path.
- Let the public reach the hospital web server (HTTP) over the internet.
- Block access to the patient database server from outside the hospital.
- Keep the BGP sessions working while the ACL is applied.
- Show that BGP sessions form, routes are exchanged, connectivity works and the ACL enforces the policy.

3. Network Topology and Devices

Figure 1 shows the topology built in Packet Tracer. The three routers are connected in a triangle. The blue one is the hospital (AS 65001), and other two are the ISPs. Every LAN has its own switch and hosts.

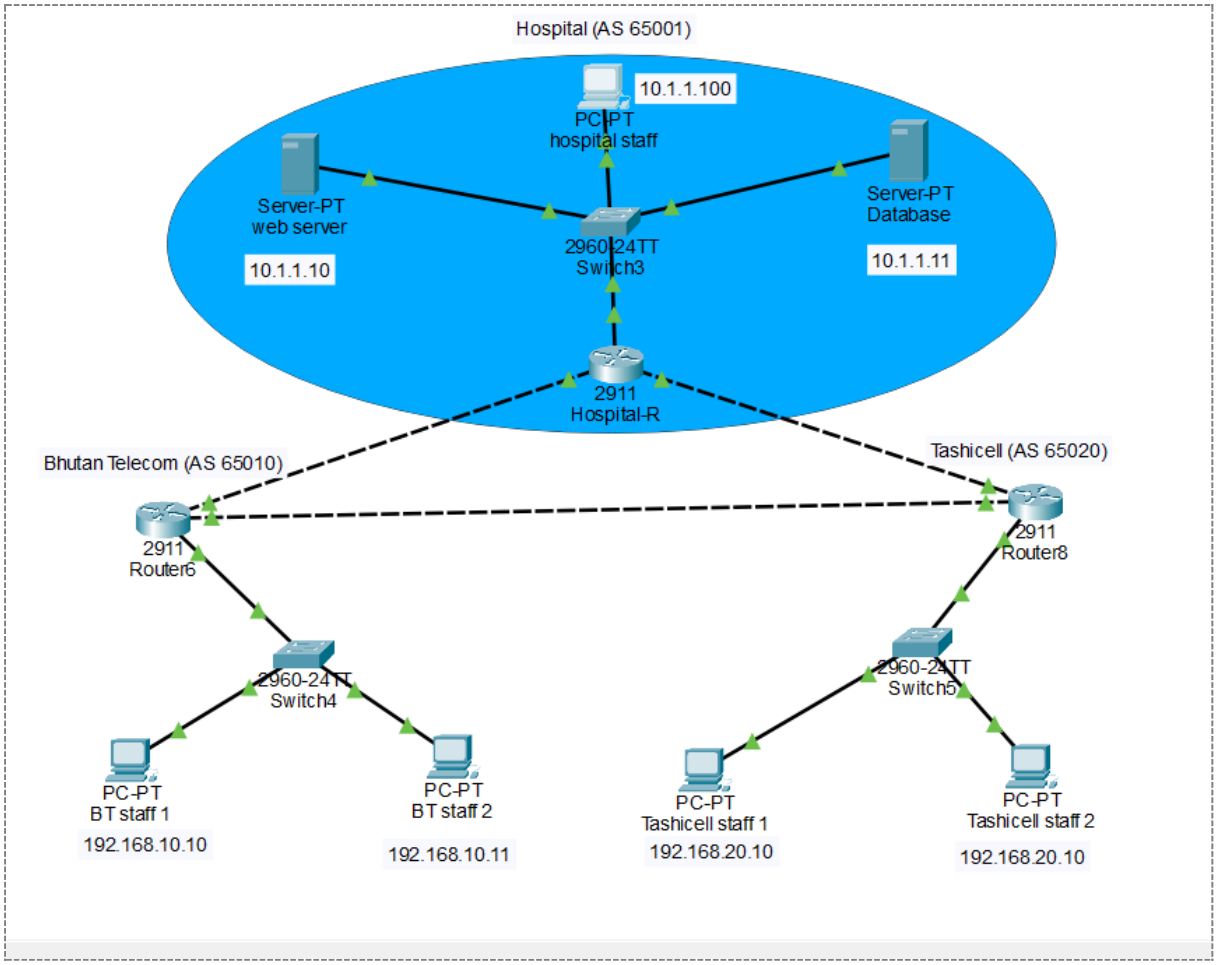


Figure 1: *Packet Tracer topology. Hospital-R (AS 65001) is linked to Router6, Bhutan Telecom (AS 65010) and Router8, Tashicell (AS 65020). Router6 and Router8 are also linked directly to each other. The hospital LAN holds the web server, the database and a staff PC behind the hospital switch.*

The table below lists every device in the topology.

Device	Type	Location	Role
Hospital-R	Router	AS 65001	Hospital edge router. Runs eBGP with Router6 and Router8 and applies the database ACL.
Router6 (BT)	Router	AS 65010	Bhutan Telecom router. Runs eBGP with Hospital-R and Router8.
Router8 (Tashicell)	Router	AS 65020	Tashicell router. Runs eBGP with Hospital-R and Router6.
Hospital Switch (ch3)	Switch	Hospital LAN	Connects the server, database and staff PC to Hospital-R.
Bhutan Telecom Switch	Switch	BT LAN	Connects BT customers to Router6.
Tashicell Switch	Switch	Tashicell LAN	Connects Tashicell customers to Router8.
Web Server	Server-PT	Hospital LAN	Public appointment portal.
Database	Server-PT	Hospital LAN	Patient records database. Must not be reachable from outside.
PC0	PC	Hospital LAN	Hospital staff test host.

Device	Type	Location	Role
PC1, PC2	PC	BT LAN	Bhutan Telecom customer test hosts (public users).
PC3, PC4	PC	Tashicell LAN	Tashicell customer test hosts (public users).

4. IP Addressing and AS Plan

4.1 Autonomous systems

Device	AS number	Router ID	Role
Hospital-R	65001	1.1.1.1	Hospital edge router
Router6 (BT)	65010	2.2.2.2	Bhutan Telecom router
Router8 (Tashicell)	65020	3.3.3.3	Tashicell router

4.2 Router links

Each router-to-router link uses its own /30 subnet from the block 172.16.0.0.

Link	Subnet	Side A	Side B
Hospital-R to Router6	172.16.1.0/30	Hospital-R Gi0/0: 172.16.1.1	Router6 Gi0/0: 172.16.1.2
Hospital-R to Router8	172.16.2.0/30	Hospital-R Gi0/1: 172.16.2.1	Router8 Gi0/0: 172.16.2.2
Router6 to Router8	172.16.3.0/30	Router6 Gi0/1: 172.16.3.1	Router8 Gi0/1: 172.16.3.2

4.3 LANs and end devices

Device	Location	IP address	Mask	Gateway	Notes
Web Server	Hospital LAN	10.1.1.10	255.255.255.0	10.1.1.1	Public portal
Database	Hospital LAN	10.1.1.11	255.255.255.0	10.1.1.1	Patient records
Hospital staff	Hospital LAN	10.1.1.100	255.255.255.0	10.1.1.1	Hospital staff
BT staff 1(PC1)	BT switch	192.168.10.10	255.255.255.0	192.168.10.1	BT customer
BT staff 2(PC2)	BT switch	192.168.10.11	255.255.255.0	192.168.10.1	BT customer
Tashicell staff1(PC3)	Tashicell switch	192.168.20.10	255.255.255.0	192.168.20.1	Tashicell customer
Tashicell staff 2(PC4)	Tashicell switch	192.168.20.11	255.255.255.0	192.168.20.1	Tashicell customer

The LAN gateways are the router interfaces: Hospital-R Gi0/2 (10.1.1.1/24), Router6 Gi0/2 (192.168.10.1/24) and Router8 Gi0/2 (192.168.20.1/24).

4.4 End-device configuration

Figure 2 shows the static IP settings of PC0, PC1, PC2, PC3, PC4, the Web Server and the Database, set in the IP Configuration window of each device.

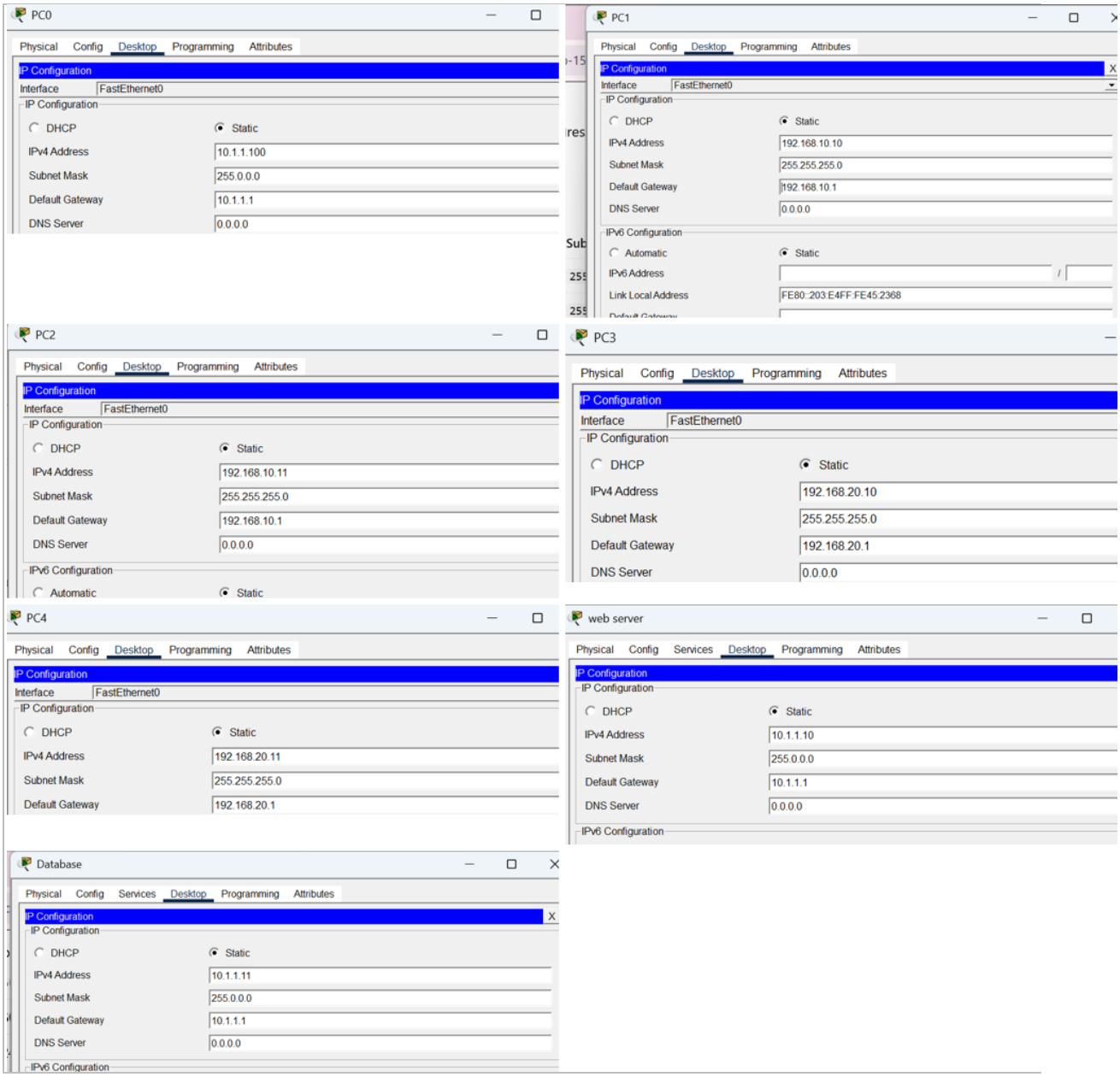


Figure 2: *Static IPv4 settings of PC0, PC1, PC2, PC3, PC4, the Web Server and the Database.*

5. Router Configuration

All three routers received the same basic hardening, then their interfaces and BGP. The sections below show the Packet Tracer console for each router. Appendix A lists the full configuration as text.

5.1 Hospital-R (hospital edge router)

Figure 3 shows the interface configuration of Hospital-R: Gi0/0 towards Bhutan Telecom (172.16.1.1/30), Gi0/1 towards Tashicell (172.16.2.1/30) and Gi0/2 for the hospital LAN (10.1.1.1/24). The LAN interface carries the PROTECT-DB ACL in the outbound direction.

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Hospital-R
Hospital-R(config)#interface gigabitEthernet 0/0
Hospital-R(config-if)# ip address 172.16.1.1 255.255.255.252
Hospital-R(config-if)# no shutdown

Hospital-R(config-if)# exit
Hospital-R(config)#interface gigabitEthernet 0/1
Hospital-R(config-if)# ip address 172.16.2.1 255.255.255.252
Hospital-R(config-if)# no shutdown

Hospital-R(config-if)# exit
Hospital-R(config)#interface gigabitEthernet 0/2
Hospital-R(config-if)# ip address 10.1.1.1 255.255.255.0
Hospital-R(config-if)# no shutdown

Hospital-R(config-if)#end
Hospital-R#write memory
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state
up

%SYS-5-CONFIG_I: Configured from console by console
```

Figure 3: *Hospital-R hostname and interfaces Gi0/0 (172.16.1.1/30), Gi0/1 (172.16.2.1/30) and Gi0/2 (10.1.1.1/24). The interfaces change state to up after no shutdown.*

Figure 4 shows the extended ACL and the BGP process. BGP runs in AS 65001 with router ID 1.1.1.1. It advertises 10.1.1.0/24 and forms eBGP neighbours with 172.16.1.2 (AS 65010) and 172.16.2.2 (AS 65020). The configuration is then saved with `write memory`.

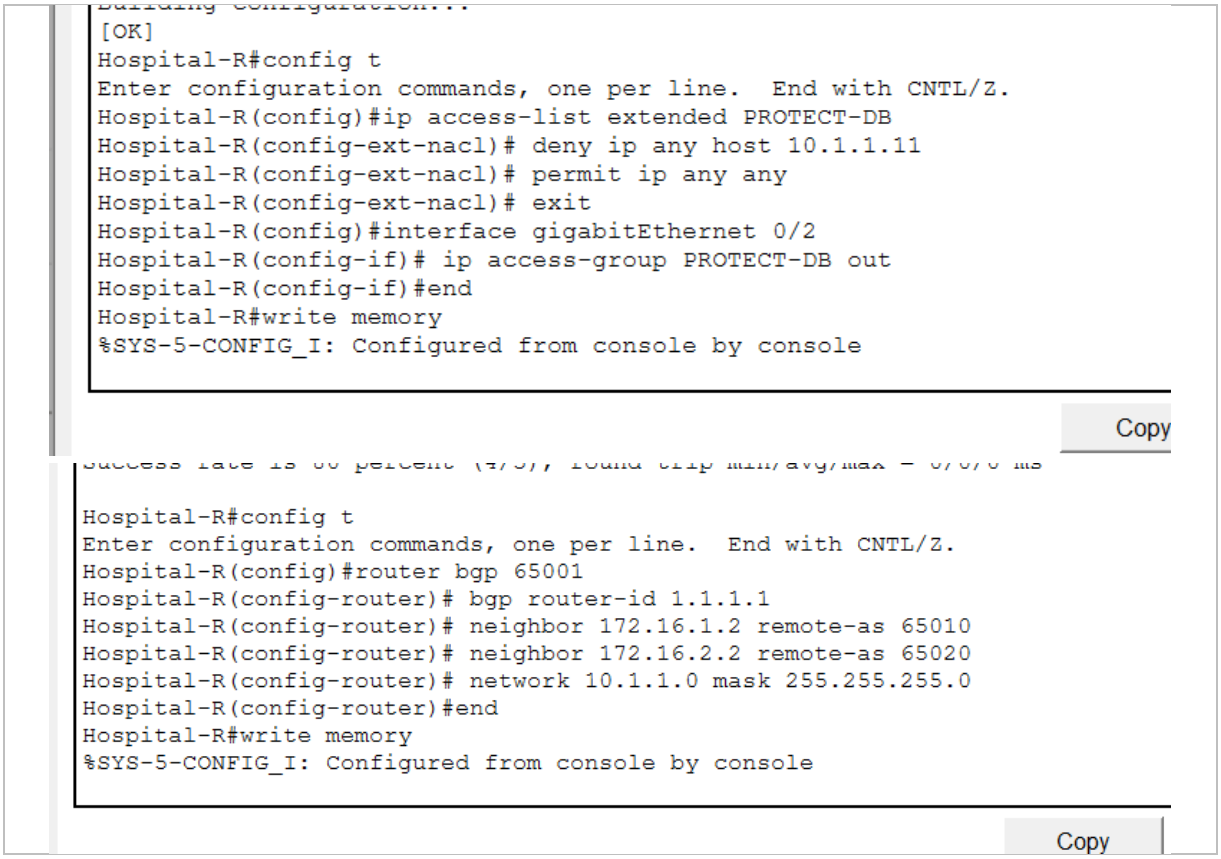


Figure 4: *Hospital-R: ACL PROTECT-DB applied outbound on Gi0/2 (top), then router bgp 65001 with its network statement and two eBGP neighbours (bottom).*

5.2 Bhutan Telecom (Router6)

Figure 5 shows the Router6 configuration. Gi0/0 is the link to Hospital-R (172.16.1.2/30), Gi0/1 is the link to Tashicell (172.16.3.1/30) and Gi0/2 is the customer LAN (192.168.10.1/24). BGP runs in AS 65010 with router ID 2.2.2.2, advertises 192.168.10.0/24 and peers with Hospital-R (172.16.1.1, AS 65001) and Router8 (172.16.3.2, AS 65020).



```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router6
Router6(config)#interface gigabitEthernet 0/0
Router6(config-if)# ip address 172.16.1.2 255.255.255.252
Router6(config-if)# no shutdown

Router6(config-if)# exit
Router6(config)#interface gigabitEthernet 0/1
Router6(config-if)# ip address 172.16.3.1 255.255.255.252
Router6(config-if)# no shutdown

Router6(config-if)# exit
Router6(config)#interface gigabitEthernet 0/2
Router6(config-if)# ip address 192.168.10.1 255.255.255.0
Router6(config-if)# no shutdown

Router6(config-if)#end
Router6#write memory
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to
up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to
up

%SYS-5-CONFIG_I: Configured from console by console
```

```
Router6#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router6(config)#router bgp 65010
Router6(config-router)# bgp router-id 2.2.2.2
Router6(config-router)# neighbor 172.16.1.1 remote-as 65001
Router6(config-router)# neighbor 172.16.3.2 remote-as 65020
Router6(config-router)# network 192.168.10.0 mask 255.255.255.0
Router6(config-router)#end
Router6#write memory
%SYS-5-CONFIG_I: Configured from console by console
%BGP-5-ADJCHANGE: neighbor 172.16.1.1 Up
```

Figure 5: *Router6 interfaces (top) and router bgp 65010 (bottom): network 192.168.10.0/24 and eBGP neighbours 172.16.1.1 (AS 65001) and 172.16.3.2 (AS 65020). The console reports neighbor 172.16.1.1 Up.*

5.3 Tashicell (Router8)

Figure 6 shows the Router8 configuration. Gi0/0 is the link to Hospital-R (172.16.2.2/30), Gi0/1 is the link to Router6 (172.16.3.2/30) and Gi0/2 is the customer LAN (192.168.20.1/24). BGP runs in AS 65020 with router ID 3.3.3.3, advertises 192.168.20.0/24 and peers with Hospital-R (172.16.2.1, AS 65001) and Router6 (172.16.3.1, AS 65010).

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router8
Router8(config)#interface gigabitEthernet 0/0
Router8(config-if)# ip address 172.16.2.2 255.255.255.252
Router8(config-if)# no shutdown

Router8(config-if)# exit
Router8(config)#interface gigabitEthernet 0/1
Router8(config-if)# ip address 172.16.3.2 255.255.255.252
Router8(config-if)# no shutdown

Router8(config-if)# exit
Router8(config)#interface gigabitEthernet 0/2
Router8(config-if)# ip address 192.168.20.1 255.255.255.0
Router8(config-if)# no shutdown

Router8(config-if)#end
Router8#write memory
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to
up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to
up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to
up

%SYS-5-CONFIG_I: Configured from console by console
```

```
Router8>enable
Router8#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router8(config)#router bgp 65020
Router8(config-router)# bgp router-id 3.3.3.3
Router8(config-router)# neighbor 172.16.2.1 remote-as 65001
Router8(config-router)# neighbor 172.16.3.1 remote-as 65010
Router8(config-router)# network 192.168.20.0 mask 255.255.255.0
Router8(config-router)#end
Router8#write memory
%SYS-5-CONFIG_I: Configured from console by console
%BGP-5-ADJCHANGE: neighbor 172.16.2.1 Up
%BGP-5-ADJCHANGE: neighbor 172.16.3.1 Up
```

Figure 6: *Router8 interfaces (top) and router bgp 65020 (bottom): network 192.168.20.0/24 and eBGP neighbours 172.16.2.1 (AS 65001) and 172.16.3.1 (AS 65010). The console reports both neighbours Up.*

5.4
Link connectivity check

Before BGP was configured, each directly connected link was tested with ping. Figure 7 shows Hospital-R pinging Router6 (172.16.1.2) and Router8 (172.16.2.2), and Router6 pinging Router8 (172.16.3.2). Each test shows a success rate of 80 percent (4 of 5). The first packet is lost while the router resolves its neighbour with ARP, which is normal, and the other four succeed.

```
[OK]
Hospital-R#ping 172.16.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Hospital-R#
```

Copy

```
up

Hospital-R#ping 172.16.2.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Hospital-R#
```

Copy

```
Router6#ping 172.16.3.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.3.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Router6#
```

Copy

Figure 7: *Ping tests on the router links: Hospital-R to 172.16.1.2 and 172.16.2.2, and Router6 to 172.16.3.2. Success rate 80 percent (the first packet is lost to ARP).*

6. BGP and eBGP Implementation

6.1 How BGP is used here

BGP is the routing protocol that connects independent networks on the internet. Each network is an Autonomous System with its own AS number. BGP exchanges prefixes (network addresses) and the AS-path, the list of ASes a route has crossed. Routers use the AS-path to choose between routes.

A BGP session between routers in **different** ASes is called **eBGP**. All three sessions in this design are eBGP because Hospital-R, Router6 and Router8 each have a different AS number. The sessions are set up with `neighbor <address> remote-as <AS>`. Each router advertises its own LAN with a `network` command, and each router learns the other two LANs from its neighbours.

Router	Local AS	Advertises	eBGP neighbours
Hospital-R	65001	10.1.1.0/24	172.16.1.2 (Router6, AS 65010); 172.16.2.2 (Router8, AS 65020)
Router6	65010	192.168.10.0/24	172.16.1.1 (Hospital-R, AS 65001); 172.16.3.2 (Router8, AS 65020)
Router8	65020	192.168.20.0/24	172.16.2.1 (Hospital-R, AS 65001); 172.16.3.1 (Router6, AS 65010)

6.2 Redundancy and path selection

Every router has two ways to reach each of the other networks. Hospital-R reaches the Bhutan Telecom LAN directly (AS-path 65010) or through Tashicell (AS-path 65020 65010). BGP prefers the shorter AS-path, so the direct link is the best path and the longer one is kept as a backup. If the direct link fails, BGP switches to the backup path without any manual change. Figure 14 shows both paths in the BGP table.

This design does not use iBGP, because each AS contains a single router. iBGP would be needed only if an AS had two or more BGP routers.

7. ACL Implementation

Hospital-R uses one named extended ACL, **PROTECT-DB**, to protect the patient database. Figure 4 shows it being configured and Figure 11 shows it in the running router.

7.1 PROTECT-DB: protecting the database

PROTECT-DB is applied **outbound** on the hospital LAN interface (Gi0/2). Packets are checked from the top down, and the first match wins.

Line	Rule	Purpose
10	deny ip any host 10.1.1.11	Blocks every packet from outside to the database (patient records).
20	permit ip any any	Allows all other traffic, including the web server. Without it the implicit deny at the end of every ACL would block the whole LAN.

7.2 Why outbound on Gi0/2

Traffic from both ISPs, whether it arrives on Gi0/0 or Gi0/1, leaves Hospital-R through Gi0/2 to reach the hospital LAN. One ACL on that interface therefore covers both paths. The ACL does not touch the BGP sessions, because they run on the link interfaces, so routing is not affected.

7.3 Limitation of PROTECT-DB

The web server, the database and PC0 share one subnet. Traffic between them goes through the switch and never passes the router, so PROTECT-DB cannot filter it. In this lab, the database is protected from the **internet** (the ISP networks), which is the stated requirement. For this reason PC0 can still reach the database.

A second effect is that the database cannot start connections towards the ISP networks and receive replies, because the replies are matched by the deny rule. This is expected and is not a routing fault.

8. Testing and Verification

Verification uses the standard Cisco show commands on each router, plus ping and browser tests from the customer PCs.

8.1 Hospital-R

show ip bgp summary. Figure 8 shows Hospital-R with router ID 1.1.1.1 in AS 65001. Both neighbours, 172.16.1.2 (AS 65010) and 172.16.2.2 (AS 65020), are listed with the number 4 in the State/PfxRcd column and uptimes of 4 minutes 22 seconds and 3 minutes 49 seconds. A number, not a word such as Active or Idle, means the session is established.

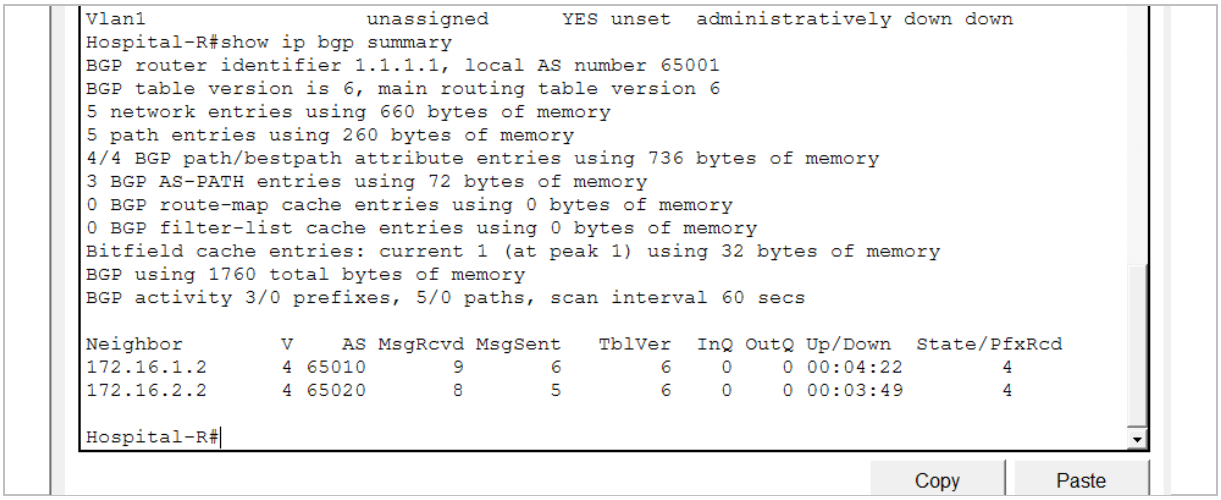


Figure 8: *show ip bgp summary on Hospital-R: neighbours 172.16.1.2 (AS 65010) and 172.16.2.2 (AS 65020).*

show ip bgp neighbors. Figure 9 should show each neighbour as an "external link" in state Established, which confirms eBGP, using BGP version 4 with the expected remote router IDs (2.2.2.2 and 3.3.3.3).

```
Hospital-R>enable
Hospital-R#show ip bgp neighbors
BGP neighbor is 172.16.1.2,  remote AS 65010, external link
  BGP version 4, remote router ID 2.2.2.2
  BGP state = Established, up for 00:01:27
  Last read 00:01:27, last write 00:01:27, hold time is 180, keepalive interval is
60 seconds
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Address family IPv4 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0


BGP neighbor is 172.16.2.2,  remote AS 65020, external link
  BGP version 4, remote router ID 3.3.3.3
  BGP state = Established, up for 00:01:27
  Last read 00:01:27, last write 00:01:27, hold time is 180, keepalive interval is
60 seconds
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Address family IPv4 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

                         Sent       Rcvd
    Opens:                  1          1
    Notifications:          0          0
    Updates:                3          3
    Keepalives:             2          2
    Route Refresh:          0          0
    Total:                  6          6
  Default minimum time between advertisements runs is 30 seconds

 For address family: IPv4 Unicast
  BGP table version 6, neighbor version 6/0
  Output queue size : 0
  Index 1, Offset 0, Mask 0x2
  1 update-group member
                                 Sent       Rcvd
  Prefix activity:               ----       ----
```

Figure 9: *show ip bgp neighbors on Hospital-R: external link, state Established.*

show ip route. Figure 10 shows the routing table of Hospital-R. Besides the connected (C) and local (L) routes, it has two BGP routes: 192.168.10.0/24 via 172.16.1.2 (Bhutan Telecom) and 192.168.20.0/24 via 172.16.2.2 (Tashicell). The code B marks BGP routes and [20/0] shows the administrative distance 20 of eBGP.

```
Hospital-R#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.1.0/24 is directly connected, GigabitEthernet0/2
L       10.1.1.1/32 is directly connected, GigabitEthernet0/2
     172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       172.16.1.0/30 is directly connected, GigabitEthernet0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0
C       172.16.2.0/30 is directly connected, GigabitEthernet0/1
L       172.16.2.1/32 is directly connected, GigabitEthernet0/1
B    192.168.10.0/24 [20/0] via 172.16.1.2, 00:00:00
B    192.168.20.0/24 [20/0] via 172.16.2.2, 00:00:00

Hospital-R#
```

Figure 10: *show ip route on Hospital-R: the BT and Tashicell LANs are learned through eBGP (code B, [20/0]).*

show access-lists. Figure 11 shows the ACL PROTECT-DB with its two lines: 10 deny ip any host 10.1.1.11 and 20 permit ip any any. This capture was taken before the blocked tests, so it shows no match counters yet. Run the command again after the tests in Section 8.4 and the counter on the deny line should rise.

```
Hospital-R#show access-lists
Extended IP access list PROTECT-DB
    10 deny ip any host 10.1.1.11
    20 permit ip any any

Hospital-R#
```

Figure 11: *show access-lists on Hospital-R: PROTECT-DB with its deny and permit lines.*

show ip interface g0/2. Figure 12 confirms where the ACL is applied: the output reads "Outgoing access list is PROTECT-DB" and "Inbound access list is not set" on Gi0/2 (10.1.1.1/24), which is up with line protocol up.

```
Hospital-R#show ip interface g0/2
GigabitEthernet0/2 is up, line protocol is up (connected)
  Internet address is 10.1.1.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is PROTECT-DB
  Inbound  access list is not set
  Proxy ARP is enabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachables are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP Fast switching turbo vector
  IP multicast fast switching is disabled
  IP multicast distributed fast switching is disabled
  Router Discovery is disabled
  IP output packet accounting is disabled
  IP access violation accounting is disabled
  TCP/IP header compression is disabled
  RTP/IP header compression is disabled
  Probe proxy name replies are disabled
  Policy routing is disabled
  Network address translation is disabled
  BGP Policy Mapping is disabled
  Input features: MCI Check
  WCCP Redirect outbound is disabled
  WCCP Redirect inbound is disabled
  WCCP Redirect exclude is disabled

Hospital-R#
```

Figure 12: *show ip interface g0/2 on Hospital-R: PROTECT-DB is applied as the outgoing access list.*

show ip interface brief. Figure 13 shows Gi0/0 (172.16.1.1), Gi0/1 (172.16.2.1) and Gi0/2 (10.1.1.1) in the up/up state. The unused Vlan1 interface is administratively down.

```
Building configuration...
[OK]
Hospital-R#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     172.16.1.1      YES manual up                    up
GigabitEthernet0/1     172.16.2.1      YES manual up                    up
GigabitEthernet0/2     10.1.1.1        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Hospital-R#
```

Copy Paste

Figure 13: *show ip interface brief on Hospital-R: Gi0/0, Gi0/1 and Gi0/2 are up/up.*

show ip bgp. Figure 14 shows the BGP table. The hospital's own prefix 10.1.1.0/24 has next hop 0.0.0.0 and weight 32768, which means it is locally originated. Each remote LAN appears twice. The best path is marked ">" and has the shorter AS-path: 192.168.10.0/24 via 172.16.1.2 (path 65010) and 192.168.20.0/24 via 172.16.2.2 (path 65020). The longer path through the other ISP is kept as the backup (65020 65010 and 65010 65020).

```
Hospital-R#show ip bgp
BGP table version is 6, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 10.1.1.0/24       0.0.0.0                  0     0 32768 i
*> 192.168.10.0/24   172.16.1.2               0     0     0 65010 i
*                    172.16.2.2               0     0     0 65020 65010 i
*> 192.168.20.0/24   172.16.2.2               0     0     0 65020 i
*                    172.16.1.2               0     0     0 65010 65020 i

Hospital-R#
```

Figure 14: *show ip bgp on Hospital-R: the local prefix, and two paths to each remote LAN. The shorter AS-path is the best path (">").*

8.2 Bhutan Telecom (Router6)

show ip bgp summary. Figure 15 should show Router6 with router ID 2.2.2.2 in AS 65010 and both neighbours, 172.16.1.1 (Hospital-R, AS 65001) and 172.16.3.2 (Router8, AS 65020), established.

```
Router6>enable
Router6#show ip bgp summary
BGP router identifier 2.2.2.2, local AS number 65010
BGP table version is 8, main routing table version 6
5 network entries using 660 bytes of memory
5 path entries using 260 bytes of memory
4/4 BGP path/bestpath attribute entries using 736 bytes of memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 1760 total bytes of memory
BGP activity 3/0 prefixes, 5/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.16.1.1      4 65001      88      81        8    0    0 01:19:24        4
172.16.3.2      4 65020      88      81        8    0    0 01:19:24        4

Router6#
```

Copy Paste

Figure 15: *show ip bgp summary on Router6: both eBGP neighbours are up.*

show ip route bgp and show ip bgp. Figure 16 should show that Router6 learned the hospital LAN 10.1.1.0/24 via 172.16.1.1 and the Tashicell LAN 192.168.20.0/24 via 172.16.3.2, plus its own prefix 192.168.10.0/24 (local, weight 32768) in the BGP table.

```
Router6#show ip route bgp
B    10.1.1.0 [20/0] via 172.16.1.1, 00:00:00
B    192.168.20.0/24 [20/0] via 172.16.3.2, 00:00:00

Router6#
```

```
Router6#show ip bgp
BGP table version is 8, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 10.1.1.0/24       172.16.1.1               0     0      0 65001 i
*                    172.16.3.2               0     0      0 65020 65001 i
*> 192.168.10.0/24   0.0.0.0                  0     0 32768 i
*> 192.168.20.0/24   172.16.3.2               0     0      0 65020 i
*                    172.16.1.1               0     0      0 65001 65020 i

Router6#
```

Figure 16: *show ip route bgp and show ip bgp on Router6.*

8.3 Tashicell (Router8)

show ip bgp summary. Figure 17 should show Router8 with router ID 3.3.3.3 in AS 65020 and both neighbours, 172.16.2.1 (Hospital-R, AS 65001) and 172.16.3.1 (Router6, AS 65010), established. With this output, all three eBGP sessions of the triangle are working.

```
Router8>enable
Router8#show ip bgp summary
BGP router identifier 3.3.3.3, local AS number 65020
BGP table version is 9, main routing table version 6
5 network entries using 660 bytes of memory
5 path entries using 260 bytes of memory
4/4 BGP path/bestpath attribute entries using 736 bytes of memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 1760 total bytes of memory
BGP activity 3/0 prefixes, 5/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.16.2.1      4 65001      88      86        9    0    0 00:19:12        4
172.16.3.1      4 65010      92      84        9    0    0 01:22:13        4

Router8#
```

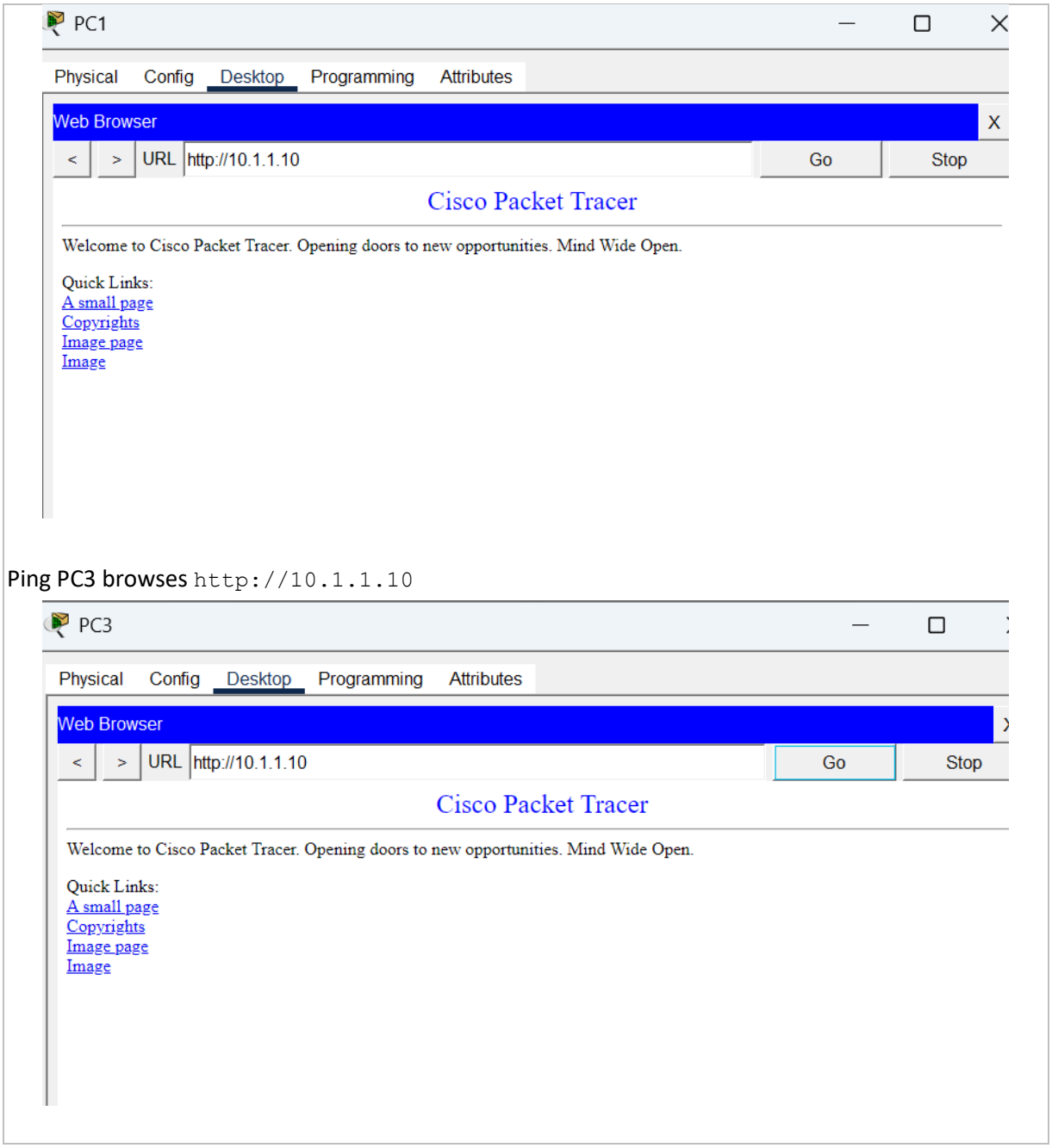
Figure 17: *show ip bgp summary on Router8: both eBGP neighbours are up.*

8.4 Connectivity and ACL tests

Allowed traffic. In Figure 18, PC1 (Bhutan Telecom) and PC3 (Tashicell) open http://10.1.1.10 in the Packet Tracer web browser. The web server page loads on both. This proves that traffic crosses the

ISP network and the hospital edge router, and that the public portal is reachable from a customer network.

Ping PC1 browses http://10.1.1.10



Ping PC3 browses http://10.1.1.10

Figure 18: *PC1 (top) and PC3 (bottom) browse to the hospital web server at 10.1.1.10 and the page loads.*

Blocked traffic. In Figure 19, a Bhutan Telecom PC and PC3 (Tashicell) ping 10.1.1.11, the database. All four packets fail on each PC (100 percent loss), and the replies "Destination host unreachable" come from Hospital-R's own ISP-facing addresses, 172.16.1.1 and 172.16.2.1. The ACL stops the request at Hospital-R, so the database is not reachable from outside the hospital. This is the required security result.

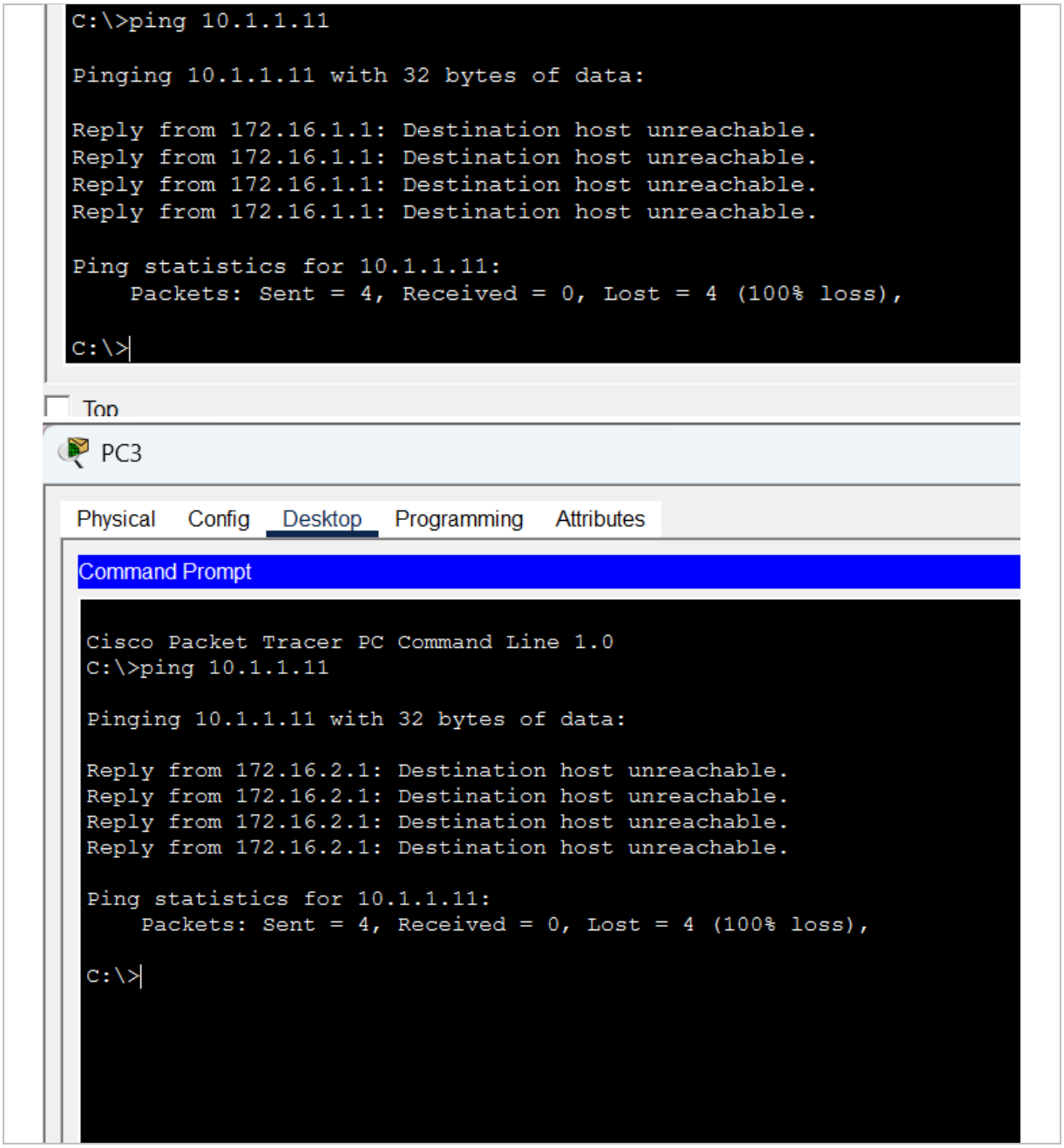
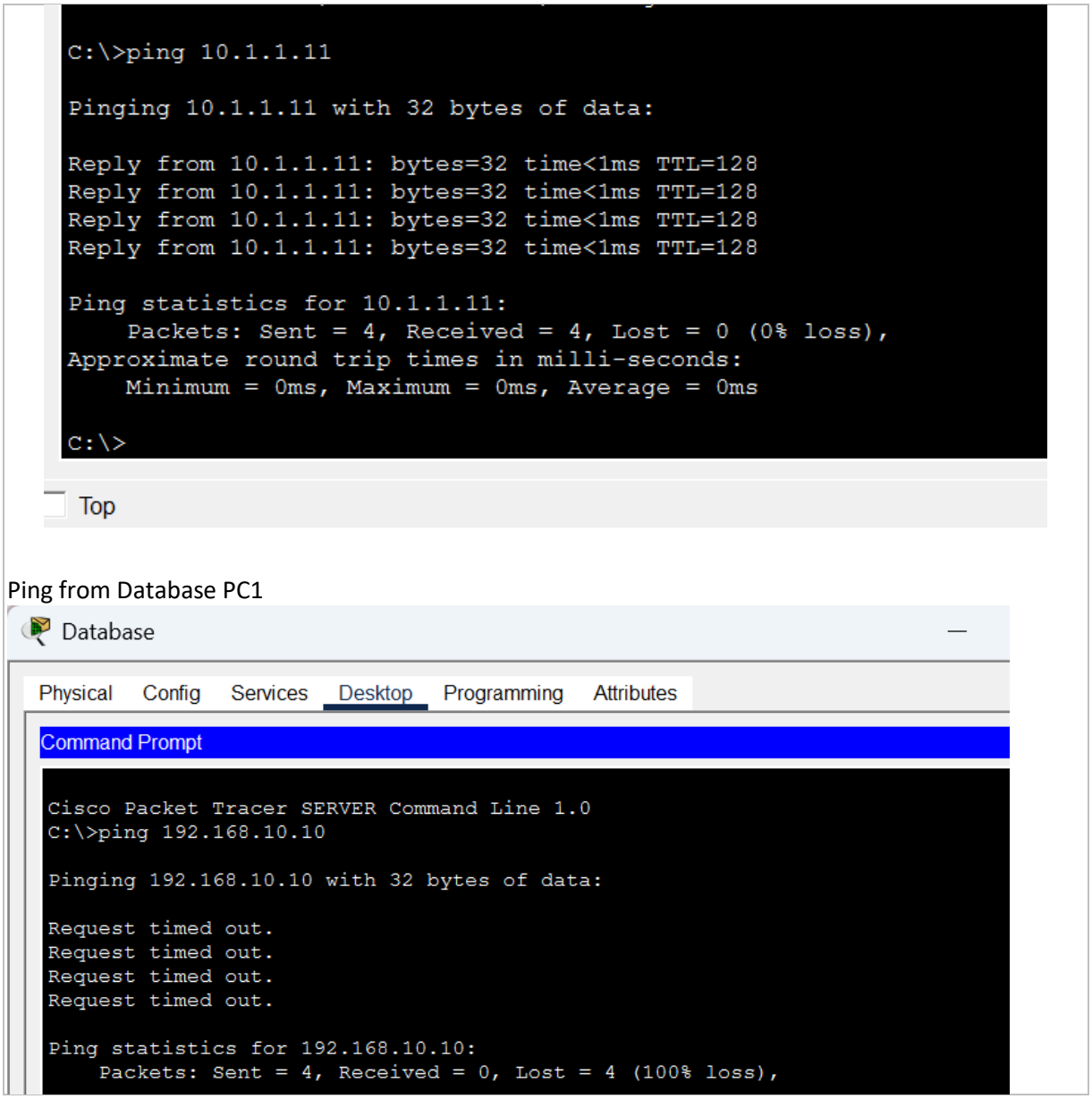


Figure 19: *Pings to the database at 10.1.1.11 from a Bhutan Telecom PC (reply from 172.16.1.1) and from PC3 (reply from 172.16.2.1): Destination host unreachable, 100 percent loss.*

Hospital-side tests. In Figure 20, PC0 on the hospital LAN pings the database and gets four replies (0 percent loss), because both are on the same LAN and the traffic never passes through Hospital-R. The database pinging 192.168.10.10 (PC1) fails with four timeouts, because the reply from PC1 is matched by the deny rule on its way back to the database.

Ping from PC0 to Database

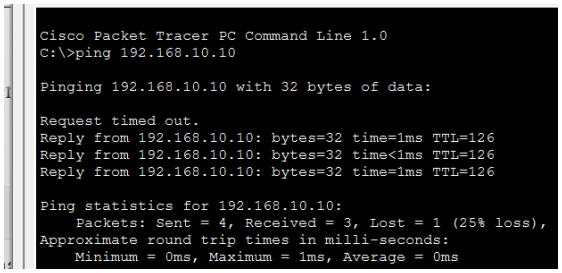


Ping from Database PC1

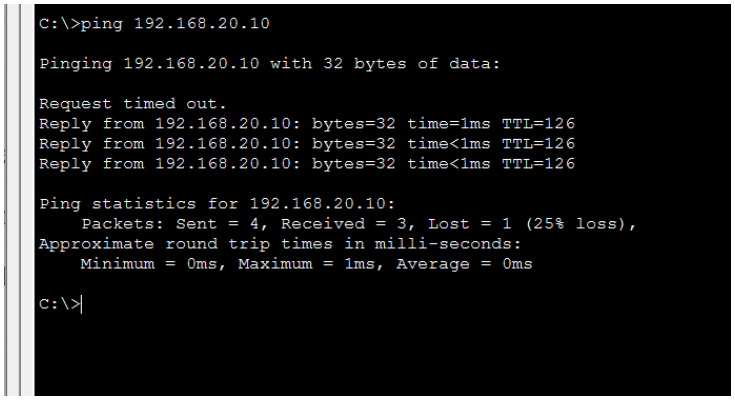
Figure 20: *PC0 pings the database (top) and succeeds. The database pings PC1 at 192.168.10.10 (bottom) and times out.*

Cross-network pings. Figure 21 shows successful pings between the three LANs: to 192.168.10.10, 192.168.20.10, 192.168.20.11 and the web server at 10.1.1.10. Each shows 3 of 4 replies, because the first packet is lost while ARP resolves. The table below lists the full set of ping tests.

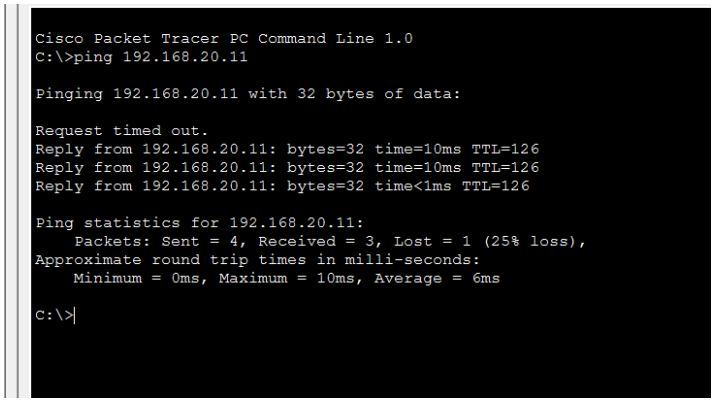
Ping from PC0 to PC1



Ping from PC0 to PC3



Ping from PC1 to PC4



Ping from PC1 to Web Server

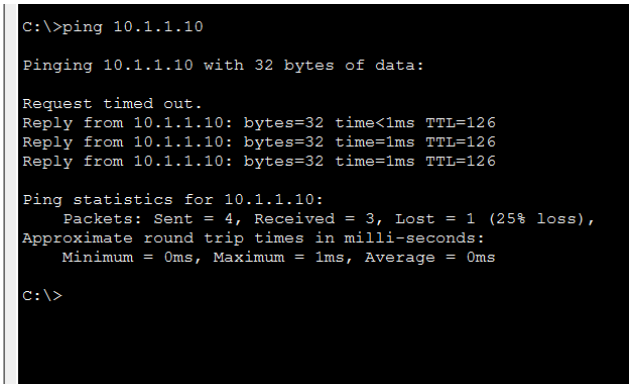


Figure 21: *Successful pings across the three LANs (the first packet of each test is lost to ARP).*

From	To	Expected	Reason
PC0	PC1 (192.168.10.10)	Success	Route learned through eBGP
PC0	PC3 (192.168.20.10)	Success	Route learned through eBGP
PC1	PC4 (192.168.20.11)	Success	Router6 and Router8 are directly linked
PC1	Web Server (10.1.1.10)	Success	The ACL denies only the database
PC0	Database (10.1.1.11)	Success	Same LAN, so the router and ACL are not involved
PC1	Database (10.1.1.11)	Fails	Denied by PROTECT-DB
PC3	Database (10.1.1.11)	Fails	Denied by PROTECT-DB
Database	PC1	Fails	The reply is matched by the deny rule

8.5 Summary of results

No.	Test	Evidence	Result
1	Hospital-R to Router6 eBGP session	Figures 8, 9	Established
2	Hospital-R to Router8 eBGP session	Figures 8, 9	Established
3	Router6 sessions to Hospital-R and Router8	Figure 15	Both up
4	Router8 sessions to Hospital-R and Router6	Figure 17	Both up
5	Hospital-R learns BT and Tashicell LANs	Figures 10, 14	Pass, with a backup path

No.	Test	Evidence	Result
6	Router6 learns hospital and Tashicell LANs	Figure 16	Pass
7	Hospital-R interfaces	Figure 13	All up/up
8	Customer reaches the web server	Figure 18	Page loads
9	Customer blocked from the database	Figures 19, 11	No reply, counters rise
10	Failover to the backup path	Figures 22, 23	Traffic uses Tashicell

9. Conclusion

The project designed and built a multihomed hospital network in Packet Tracer. Three routers, each in its own Autonomous System, form a triangle of eBGP sessions. The verification steps show how to confirm that all sessions are established, that each router learns the other LANs, and that BGP keeps a backup path through the other provider. An extended ACL protects the patient database: a customer can open the web portal but cannot reach the database.

full Configuration

Passwords are shown as <password>.

A.1 Hospital-R

```
enable
configure terminal
hostname Hospital-R
no ip domain-lookup
enable secret <password>
service password-encryption
banner motd #Authorized access only#
line console 0
 password <password>
 login
 exit
line vty 0 4
 password <password>
 login
 exit
interface GigabitEthernet0/0
 description To Bhutan Telecom
 ip address 172.16.1.1 255.255.255.252
 no shutdown
 exit
interface GigabitEthernet0/1
 description To Tashicell
 ip address 172.16.2.1 255.255.255.252
 no shutdown
 exit
interface GigabitEthernet0/2
 description Hospital LAN
 ip address 10.1.1.1 255.255.255.0
 ip access-group PROTECT-DB out
 no shutdown
 exit
ip access-list extended PROTECT-DB
 deny ip any host 10.1.1.11
 permit ip any any
 exit
router bgp 65001
 bgp router-id 1.1.1.1
 network 10.1.1.0 mask 255.255.255.0
 neighbor 172.16.1.2 remote-as 65010
 neighbor 172.16.2.2 remote-as 65020
 exit
end
write memory
```

A.2 Bhutan Telecom (Router6)

```
enable
configure terminal
hostname Router6
no ip domain-lookup
enable secret <password>
service password-encryption
banner motd #Authorized access only#
line console 0
```

```
 password <password>
 login
 exit
line vty 0 4
 password <password>
 login
 exit
interface GigabitEthernet0/0
 description To Hospital-R
 ip address 172.16.1.2 255.255.255.252
 no shutdown
 exit
interface GigabitEthernet0/1
 description To Tashicell
 ip address 172.16.3.1 255.255.255.252
 no shutdown
 exit
interface GigabitEthernet0/2
 description BT customer LAN
 ip address 192.168.10.1 255.255.255.0
 no shutdown
 exit
router bgp 65010
 bgp router-id 2.2.2.2
 network 192.168.10.0 mask 255.255.255.0
 neighbor 172.16.1.1 remote-as 65001
 neighbor 172.16.3.2 remote-as 65020
 exit
end
write memory
```

A.3 Tashicell (Router8)

```
enable
configure terminal
hostname Router8
no ip domain-lookup
enable secret <password>
service password-encryption
banner motd #Authorized access only#
line console 0
 password <password>
 login
 exit
line vty 0 4
 password <password>
 login
 exit
interface GigabitEthernet0/0
 description To Hospital-R
 ip address 172.16.2.2 255.255.255.252
 no shutdown
 exit
interface GigabitEthernet0/1
 description To Bhutan Telecom
 ip address 172.16.3.2 255.255.255.252
 no shutdown
 exit
interface GigabitEthernet0/2
 description Tashicell customer LAN
 ip address 192.168.20.1 255.255.255.0
```

```
 no shutdown
 exit
router bgp 65020
 bgp router-id 3.3.3.3
 network 192.168.20.0 mask 255.255.255.0
 neighbor 172.16.2.1 remote-as 65001
 neighbor 172.16.3.1 remote-as 65010
 exit
end
write memory
```

Marking criteria

Sl.no	Assessment criteria	Marks	Remarks
1	Network Scenario and topology design=1		
2	BPG implementation(eBPG)=3		
3	ACL Implementation=1		
4	Testing and Verification=1		
5	Report=2		
6	Viva voce=2		
Total marks=10			